

Past paper 2018. MATH-201

(i) Basis of Column space: $\begin{bmatrix} 1 & 2 & 2 & -1 \\ 3 & 6 & 5 & 0 \\ 1 & 2 & 1 & 2 \end{bmatrix}$

$$A = \begin{bmatrix} 1 & 2 & 2 & -1 \\ 3 & 6 & 5 & 0 \\ 1 & 2 & 1 & 2 \end{bmatrix} \rightarrow R \begin{bmatrix} 1 & 2 & 2 & -1 \\ 0 & 0 & -1 & 3 \\ 0 & 0 & -1 & 3 \end{bmatrix} \begin{matrix} R_2 - 3R_1 \\ R_3 - R_1 \end{matrix} \rightarrow R \begin{bmatrix} 1 & 2 & 2 & -1 \\ 0 & 0 & -1 & 3 \\ 0 & 0 & -1 & 3 \end{bmatrix} \begin{matrix} \\ \\ R_3 - R_2 \end{matrix}$$

$$R \begin{bmatrix} 1 & 2 & 2 & -1 \\ 0 & 0 & -1 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{matrix} \\ R_3 + R_2 \\ \end{matrix} \text{ reduced echelon form. } R = \begin{bmatrix} 1 & 2 & 2 & -1 \\ 0 & 0 & -1 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

leading 1 waly columns

C_1 and C_3 from original matrix
 $e_1 = \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}$ and $e_3 = \begin{bmatrix} 2 \\ 5 \\ 1 \end{bmatrix}$ form basis for column space of given matrix

Basis of Row space = $r_1 = [1 \ 2 \ 2 \ -1]$, $r_2 = [0 \ 0 \ 1 \ 3]$

Dimension of row space = Rank of A

$\rightarrow$ Nullity (A) + Rank (A) = n

$$N = 3 - 2 = 1$$

(ii) If A is nonsingular inverse is $\begin{bmatrix} 2 & 1 \\ 4 & 1 \end{bmatrix}$, find A

$$A^{-1} = \begin{bmatrix} 2 & 1 \\ 4 & 1 \end{bmatrix} \quad (A^{-1})^{-1} = \frac{\text{Adj } A^{-1}}{|A^{-1}|} \quad |A^{-1}| = 2 - 4 = -2$$

$$\text{Adj } A^{-1} = \begin{bmatrix} 1 & -1 \\ -4 & 2 \end{bmatrix} \quad (A^{-1})^{-1} = \frac{1}{-2} \begin{bmatrix} 1 & -1 \\ -4 & 2 \end{bmatrix} = A$$

(iii) $\begin{bmatrix} 1 & -2 & 3 & -1 \\ 2 & -1 & 2 & 2 \\ 3 & 1 & 2 & 3 \end{bmatrix}$ into reduced echelon form.

$$R \begin{bmatrix} 1 & -2 & 3 & -1 \\ 0 & 3 & -4 & 4 \\ 0 & 7 & -7 & 6 \end{bmatrix} \begin{matrix} R_2 - 2R_1 \\ R_3 - 3R_1 \end{matrix} \quad R \begin{bmatrix} 1 & -2 & 3 & -1 \\ 0 & 1 & -4/3 & 4/3 \\ 0 & 7 & -7 & 6 \end{bmatrix} \begin{matrix} \\ \\ R_3 - 7R_2 \end{matrix} = R \begin{bmatrix} 1 & -2 & 3 & -1 \\ 0 & 1 & -4/3 & 4/3 \\ 0 & 0 & 7/3 & -10/3 \end{bmatrix}$$

$$R \begin{bmatrix} 1 & -2 & 3 & -1 \\ 0 & 1 & -4/3 & 4/3 \\ 0 & 0 & 1 & -10/7 \end{bmatrix} \begin{matrix} \\ \\ 3R_3 \end{matrix} \quad R \begin{bmatrix} 1 & -2 & 0 & 23/7 \\ 0 & 1 & 0 & -4/7 \\ 0 & 0 & 1 & -10/7 \end{bmatrix} \begin{matrix} \\ R_2 + 4/3 R_3 \\ R_1 - 3R_3 \end{matrix} = R \begin{bmatrix} 1 & 0 & 0 & 5/7 \\ 0 & 1 & 0 & -4/7 \\ 0 & 0 & 1 & -10/7 \end{bmatrix}$$

$$(iv) \begin{vmatrix} (b+c)^2 & a^2 & a^2 \\ b^2 & (c+a)^2 & b^2 \\ c^2 & c^2 & (a+b)^2 \end{vmatrix} = 2abc(a+b+c)^3$$

Taking $(a+b+c)$ common from C_1, C_2 both

$$\begin{pmatrix} 3/4 & -1/4 & 0 \\ 1/2 & 0 & 0 \\ -1/4 & 1/4 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1/2 & 1/4 & -2/4 \\ 0 & -1/2 & 0 \\ -1/2 & -1/4 & 1/4 \end{pmatrix}$$

$$A_{11} = \begin{vmatrix} -2 & 0 \\ -2 & 2 \end{vmatrix} = -4 \quad A_{21} = \begin{vmatrix} 2 & -3 \\ -2 & 2 \end{vmatrix} = 4-6=-2 \quad A_{31} = \begin{vmatrix} 2 & -3 \\ -2 & 0 \end{vmatrix} = -6$$

$$A_{12} = \begin{vmatrix} 0 & 0 \\ -2 & 2 \end{vmatrix} = 0 \quad A_{22} = \begin{vmatrix} 1 & -3 \\ -2 & 2 \end{vmatrix} = 2-6=-4 \quad A_{32} = \begin{vmatrix} 1 & -3 \\ 0 & 0 \end{vmatrix} = 0$$

$$A_{13} = \begin{vmatrix} 0 & -2 \\ -2 & -2 \end{vmatrix} = -4 \quad A_{23} = \begin{vmatrix} 1 & 2 \\ -2 & -2 \end{vmatrix} = -2+4=2 \quad A_{33} = \begin{vmatrix} 1 & 2 \\ 0 & -2 \end{vmatrix} = -2$$

$$\text{Adj } A = \begin{pmatrix} -4 & -0 & -4 \\ -(-2) & -4 & -2 \\ -6 & -(0) & -2 \end{pmatrix}^T = \begin{pmatrix} -4 & 0 & -4 \\ 2 & -4 & -2 \\ -6 & 0 & -2 \end{pmatrix}^T$$

$$= \begin{pmatrix} -4 & 2 & -6 \\ 0 & -4 & 0 \\ -4 & -2 & -2 \end{pmatrix}$$

$$A^{-1} = \begin{pmatrix} 1 & -1/2 & 3/2 \\ 0 & 1 & 0 \\ 1 & 1/2 & 1/2 \end{pmatrix}$$

4. $v_1(1, -2, 3)$, $v_2(5, 6, -1)$, $v_3(3, 2, 1)$ linearly independent?

$$av_1 + bv_2 + cv_3 = (0, 0, 0) \quad \text{where } a, b, c \in \mathbb{R}$$

$$a(1, -2, 3) + b(5, 6, -1) + c(3, 2, 1) = (0, 0, 0)$$

$$(a+5b+3c, -2a+6b+2c, 3a-b+c) = (0, 0, 0)$$

$$a+5b+3c=0 \quad -2a+6b+2c=0 \quad 3a-b+c=0$$

$$A = \begin{pmatrix} 1 & 5 & 3 \\ -2 & 6 & 2 \\ 3 & -1 & 1 \end{pmatrix} \quad B = \begin{pmatrix} 1 & 5 & 3 \\ 0 & 16 & 8 \\ 0 & -16 & -8 \end{pmatrix} \begin{matrix} R_2+2R_1 \\ R_3-3R_1 \end{matrix} \quad C = \begin{pmatrix} 1 & 5 & 3 \\ 0 & 1 & 1/2 \\ 0 & -16 & -8 \end{pmatrix} \begin{matrix} \\ \\ \frac{1}{16}R_2 \end{matrix}$$

$$B = \begin{pmatrix} 1 & 5 & 3 \\ 0 & 1 & 1/2 \\ 0 & 0 & 0 \end{pmatrix} \begin{matrix} R_3+16R_2 \\ \end{matrix} \quad \text{Not linearly independent}$$

As $|A| = 0$, So it is

linearly dependent

$$1(6+2) - 5(-2-6) + 3(-12)$$

$$8 - 5(-8) + 3(-16)$$

$$8 + 40 - 48$$

$$0 - 40 = 0$$

5. $\{(1, 2, -1), (0, 3, 1), (1, -5, 3)\}$ is basis of R^3 ?

$$a(1, 2, -1) + b(0, 3, 1) + c(1, -5, 3) = (x, y, z)$$

$$a + c, 2a + 3b - 5c, -a + b + 3c = (x, y, z)$$

$$\begin{pmatrix} 1 & 0 & 1 \\ 2 & 3 & -5 \\ -1 & 1 & 3 \end{pmatrix} \xrightarrow{R_2 - 2R_1, R_3 + R_1} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 3 & -3 \\ 0 & 1 & 4 \end{pmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 4 \\ 0 & 3 & -3 \end{pmatrix}$$

$$\xrightarrow{R_3 - 3R_2} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 4 \\ 0 & 0 & -15 \end{pmatrix}$$

$c = 0 = 0 = 0$ These vectors are basis of R^3

6. $\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$ find $P^{-1}AP$

$$A - \lambda I = \begin{pmatrix} 1-\lambda & 2 \\ 2 & 1-\lambda \end{pmatrix} = (1-\lambda)(1-\lambda) - 4 = 1 - 2\lambda + \lambda^2 - 4 = \lambda^2 - 2\lambda - 3$$

Characteristic Polynomial

$$\lambda^2 - 3\lambda + \lambda - 3 = \lambda(\lambda - 3) + 1(\lambda - 3) = (\lambda - 3)(\lambda + 1)$$

$\lambda = 3, -1$ Eigen values

$$A - 3I = \begin{pmatrix} -2 & 2 \\ 2 & -2 \end{pmatrix} \xrightarrow{\div 2} \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} \xrightarrow{R_2 + R_1} \begin{pmatrix} -1 & 1 \\ 0 & 0 \end{pmatrix} \begin{matrix} x_1 = x_2 \\ 0 \end{matrix}$$

$$\begin{pmatrix} x_1 \\ x_1 \end{pmatrix} = x_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad \left[\frac{1}{\sqrt{2}} \quad \frac{1}{\sqrt{2}} \right]^T \text{ Eigen vector. We normalize it}$$

$$A + I = \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix} \xrightarrow{\div 2} \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} \begin{matrix} x_1 = -x_2 \\ 0 \end{matrix} \Rightarrow \begin{pmatrix} -x_1 \\ -x_1 \end{pmatrix} = x_1 \begin{pmatrix} -1 \\ -1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & -1 \end{pmatrix}^T \quad \left[\frac{1}{\sqrt{2}} \quad -\frac{1}{\sqrt{2}} \right]^T$$

$$P = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix} \quad P^{-1} = P^T = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix}$$

$$P^{-1}AP = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 3 & -1 \\ 3 & 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 6 & 0 \\ 0 & -2 \end{pmatrix} = \begin{pmatrix} 3 & 0 \\ 0 & -1 \end{pmatrix}$$

$$\begin{aligned} 7. \quad & x+y+7z = -7 \\ & 2x+3y+17z = -16 \\ & x+2y+(a^2+1)z = 3a \end{aligned}$$

Determine values of a for which system of linear equation has no solution, exactly one solution and infinitely many solution.

$$\left[\begin{array}{ccc|c} 1 & 1 & 7 & -7 \\ 2 & 3 & 17 & -16 \\ 1 & 2 & a^2+1 & 3a \end{array} \right]$$

$$\begin{aligned} R_2 - 2R_1, \quad R_3 - R_1 \\ \left[\begin{array}{ccc|c} 1 & 1 & 7 & -7 \\ 0 & 1 & 3 & -2 \\ 0 & 1 & a^2-6 & 3a+7 \end{array} \right] \end{aligned}$$

$$\begin{aligned} R_3 - R_2 \\ \left[\begin{array}{ccc|c} 1 & 1 & 7 & -7 \\ 0 & 1 & 3 & -2 \\ 0 & 0 & a^2-9 & 3a+9 \end{array} \right] \end{aligned}$$

No solutions if

$$a^2-9=0 \quad 3a+9 \neq 0$$

$$a = \pm 3$$

$$a \neq -3$$

One solution if

$$a^2-9 \neq 0 \Rightarrow a \neq \pm 3 \text{ and } 3a+9 \text{ is real number}$$

Many solution if

$$a = \pm 3 \quad a = -3 \Rightarrow [0 \ 0 \ 0 \ 0]$$

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i) Decide yes..... If A is 2×2 matrix such that $A^2=0$, A must be null matrix

No

$$\text{Counter example: } A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \Rightarrow A^2 = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$A^2=0$ but A is not null matrix

In this case, A is nilpotent matrix, meaning some power of A (in this case, square) is zero matrix but A itself is not zero matrix

(ii) $v = (2, -5, 3)$ as linear combination $u_1 = (1, -3, 2)$, $u_2 = (2, -4, -1)$, $u_3 = (1, -5, 7)$

$$v = au_1 + bu_2 + cu_3$$

$$(2, -5, 3) = a(1, -3, 2) + b(2, -4, -1) + c(1, -5, 7)$$

$$(2, 5, 3) = (a+2b+c, -3a-4b-5c, 2a-b+7c)$$

$$= \begin{bmatrix} 1 & 2 & 1 & 2 \\ -3 & -4 & -5 & -5 \\ 2 & -1 & 7 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 1 & 2 \\ 0 & 2 & -2 & 1 \\ 0 & -5 & 5 & -1 \end{bmatrix} \begin{array}{l} R_2 + 3R_1 \\ R_3 - 2R_1 \end{array}$$

$$= \begin{bmatrix} 1 & 2 & 1 & 2 \\ 0 & 1 & -1 & 1/2 \\ 0 & -5 & 5 & -1 \end{bmatrix} \xrightarrow{\frac{1}{2}R_2} \begin{bmatrix} 1 & 2 & 1 & 2 \\ 0 & 1 & -1 & 1/2 \\ 0 & 0 & 0 & 3/2 \end{bmatrix} \begin{array}{l} R_3 + 5R_2 \end{array}$$

Rank of A $\neq$ Rank of Ab

Subspace or not?

$$(iii) \quad V = \{f: R \rightarrow R\} \quad W = \{f \in V : f(1) = 0\}$$

$$V = \{f: R \rightarrow R\} \quad W = \{f \in V : f(1) = 0\}$$

$$(i) \quad 0(1) = 0 \rightarrow 0 \in W \rightarrow 0 \neq W$$

$$(ii) \quad f, g \in W \quad f(1) = 0 \quad g(1) = 0$$

$$(f+g)(1) = f(1) + g(1) = 0 + 0 = 0 \rightarrow f+g \in W$$

$$(iii) \quad \alpha \in R$$

$$(\alpha f)(1) = \alpha f(1) = \alpha(0) = 0 \rightarrow \alpha f \in W$$

W is Subspace of V

$$(iv) \quad \begin{vmatrix} a & 1 & 1 & 1 \\ 1 & a & 1 & 1 \\ 1 & 1 & a & 1 \\ 1 & 1 & 1 & a \end{vmatrix} = (a-1)^3(a+3)$$

$$\begin{vmatrix} a+3 & 1 & 1 & 1 \\ a+3 & a & 1 & 1 \\ a+3 & 1 & a & 1 \\ a+3 & 1 & 1 & a \end{vmatrix} \xrightarrow{C_1 + (C_2 + C_3 + C_4) = (a+3)} \begin{vmatrix} 1 & 1 & 1 & 1 \\ 1 & a & 1 & 1 \\ 1 & 1 & a & 1 \\ 1 & 1 & 1 & a \end{vmatrix}$$

$$= (a+3) \begin{vmatrix} 1 & 1 & 1 & 1 \\ 0 & a-1 & 0 & 0 \\ 0 & 0 & a-1 & 0 \\ 0 & 0 & 0 & a-1 \end{vmatrix} = (a+3)(a-1)^3$$

v) If A and B are invertible matrices of same order
 AB is invertible then show $(AB)^{-1} = B^{-1}A^{-1}$

$$(AB)^{-1} = B^{-1}A^{-1}$$

$$1. (AB)(A^{-1}A^{-1}) = A(BB^{-1})A^{-1} \quad \text{Associative law}$$

$$= AIA^{-1} = I$$

$$= BB^{-1} = I, AA^{-1} = I$$

Similarly,

$$(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B$$

$$= B^{-1}IB = I$$

$$\text{So, } (AB)(B^{-1}A^{-1}) = (B^{-1}A^{-1})(AB)$$

$$(AB)^{-1} = B^{-1}A^{-1}$$

This proof assumes that A, B are invertible matrices of same order, ~~same number of rows and columns.~~

$$3. \quad x_1 - 2x_2 + 3x_3 = 3$$

$$2x_1 + x_2 - x_3 = 9$$

$$-3x_1 - 4x_2 + 5x_3 = -15$$

$$\begin{bmatrix} 1 & -2 & 3 & 3 \\ 2 & 1 & -1 & 9 \\ -3 & -4 & 5 & -15 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -2 & 3 & 3 \\ 0 & 5 & -7 & 3 \\ 0 & -10 & 14 & -6 \end{bmatrix} \begin{array}{l} \\ R_2 - 2R_1 \\ R_3 + 3R_1 \end{array}$$

$$\begin{bmatrix} 1 & -2 & 3 & 3 \\ 0 & 1 & -7/5 & 3/5 \\ 0 & -10 & 14 & -6 \end{bmatrix} \begin{array}{l} \\ \\ \frac{1}{5}R_2 \end{array}$$

$$\begin{bmatrix} 1 & -2 & 3 & 3 \\ 0 & 1 & -7/5 & 3/5 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{array}{l} \\ R_3 + 10R_2 \\ \end{array}$$

Rank of $A \neq$ Rank of Ab

$$4. \quad T(x_1, x_2, x_3) = (-x_3, x_1, x_1 + x_3)$$

$$\text{let } v_1 = (x_1, x_2, x_3) \quad v_2 = (y_1, y_2, y_3)$$

$$Tv_1 = T(-x_3, x_1, x_1 + x_3)$$

$$\text{for } N(T) \Rightarrow Tv = 0$$

$$x_3 = 0 \quad x_1 = 0 \quad x_1 + x_3 = 0$$

$0 = 0$ satisfied

$$N(T) = \{(x_1, x_2, x_3) \in \mathbb{R}^3 : (0, x_2, 0)\}$$

It is not one-to-one

5. $\begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix} \quad A^2 - 4A - 5I = ?$

$A^2 = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 7 & 6 & 6 \\ 8 & 9 & 6 \\ 8 & 8 & 7 \end{bmatrix}$

$4A = \begin{bmatrix} 4 & 8 & 4 \\ 8 & 4 & 8 \\ 8 & 4 & 4 \end{bmatrix} \quad 5I = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix}$

$A^2 - 4A - 5I = \begin{bmatrix} 7 & 6 & 6 \\ 8 & 9 & 6 \\ 8 & 8 & 7 \end{bmatrix} - \begin{bmatrix} 4 & 8 & 4 \\ 8 & 4 & 8 \\ 8 & 4 & 4 \end{bmatrix} - \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix}$

$= \begin{bmatrix} 3 & -2 & 2 \\ 0 & 5 & -2 \\ 0 & 4 & 2 \end{bmatrix} \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix} = \begin{bmatrix} -2 & -2 & 2 \\ 0 & 0 & -2 \\ 0 & 0 & -2 \end{bmatrix}$

6. $\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} \quad P^{-1}AP = ? \quad \text{(Done)}$

7. $M = \begin{bmatrix} 4 & 7 \\ 7 & 9 \end{bmatrix} \quad A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \quad C = \begin{bmatrix} 1 & 1 \\ 4 & 5 \end{bmatrix}$

$M = CA + GB + C_3C$

$\begin{bmatrix} 4 & 7 \\ 7 & 9 \end{bmatrix} = C_1 \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} + C_2 \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} + C_3 \begin{bmatrix} 1 & 1 \\ 4 & 5 \end{bmatrix}$

$4 = C_1 + C_2 + C_3$

$7 = C_1 + 2C_2 + C_3$

$7 = C_1 + 3C_2 + 4C_3$

$9 = C_1 + 4C_2 + 5C_3$

$\begin{bmatrix} 1 & 1 & 1 & 4 \\ 1 & 2 & 1 & 7 \\ 1 & 3 & 4 & 7 \\ 1 & 4 & 5 & 9 \end{bmatrix}$

$\sim B \begin{bmatrix} 1 & 1 & 1 & 4 \\ 0 & 1 & 0 & 3 \\ 0 & 2 & 3 & 3 \\ 0 & 3 & 4 & 5 \end{bmatrix} \begin{array}{l} R_2 - R_1 \\ R_3 - R_1 \\ R_4 - R_1 \end{array}$

$\sim B \begin{bmatrix} 1 & 1 & 1 & 4 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 3 & -3 \\ 0 & 0 & 4 & -10 \end{bmatrix} \begin{array}{l} R_3 - 2R_2 \\ R_4 - 3R_2 \end{array}$

$$R \begin{bmatrix} 1 & 1 & 1 & 4 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 4 & -10 \end{bmatrix} \xrightarrow{R_4 - 4R_3} \begin{bmatrix} 1 & 1 & 1 & 4 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & -6 \end{bmatrix}$$

$$R \begin{bmatrix} 1 & 1 & 1 & 4 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & -6 \end{bmatrix} \xrightarrow{R_1 - R_2} \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & -6 \end{bmatrix}$$

$$C_1 = 1, C_2 = 3, C_1 + C_2 + C_3 = 4 \Rightarrow C_1 = 4 + 1 - 3 = 2$$

Verification:-

$$\begin{bmatrix} 4 & 7 \\ 7 & 9 \end{bmatrix} = 2 \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} + 3 \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix} + \begin{bmatrix} 2 & 2 \\ 2 & 5 \end{bmatrix} = \begin{bmatrix} 4 & 7 \\ 7 & 9 \end{bmatrix} \quad \text{Satisfied}$$

2019 MATH-201.

(i) $f(t) = \cos t, g(t) = \sin t, h(t) = t$ are l.i.

$$a \cos t + b \sin t + ct = 0 \quad a, b, c \in \mathbb{R}$$

$$\text{Diff w.r.t } t \quad -a \sin t + b \cos t + c = 0$$

$$\text{Again diff. w.r.t } t \quad -a \cos t - b \sin t = 0$$

$$\text{Put } t = 0 \quad a = 0 \quad b = 0 \quad c = 0 \quad \text{l.i.}$$

$$(ii) \begin{bmatrix} b+c & b & c \\ c & c+a & a \\ b & a & a+b \end{bmatrix} = 2a(b^2 + c^2)$$

$$= (b+c) \{ (c+a)(a+b) - a^2 \} - b(ac+bc-ab) + c(ac-bc-ab)$$

$$= (b+c) \{ ac+bc+ab-a^2 \} - abc - b^2c + ab^2 + ac^2 - bc^2 - abc$$

$$= abc + b^2c + ab^2 + ac^2 + bc^2 + abc - 2abc - b^2c + ab^2 + ac^2 - bc^2$$

$$= 2ab^2 + 2ac^2$$

$$= 2a(b^2 + c^2)$$

$$(iii) \begin{bmatrix} 0 & 1 & -1 \\ 4 & -3 & 4 \\ 3 & -3 & 4 \end{bmatrix} \text{ is involutory}$$

$$A^2 = \begin{bmatrix} 0 & 1 & -1 \\ 4 & -3 & 4 \\ 3 & -3 & 4 \end{bmatrix} \begin{bmatrix} 0 & 1 & -1 \\ 4 & -3 & 4 \\ 3 & -3 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I \quad \text{So Given matrix is involutory}$$

iv) $V = \mathbb{R}^2$ and W is set of 1st quad of $\mathbb{R}^2$

1. $W \neq \emptyset$

$(0,0), (1,1), (2,2) \in \mathbb{R}^2$

2. $u \in W, v \in W \implies u+v \in W$

as $(1,1) + (2,-3) = (3,-2)$ which $\notin W$ as it lies in 2nd quad

3. $a \in \mathbb{R} \implies au \in W$ as

$-1(1,1) = (-1,-1) \notin W$

Not subspace of V

(v) $\begin{pmatrix} 1 & 2 & 2 & -1 \\ 3 & 6 & 5 & 0 \\ 1 & 2 & 1 & 0 \end{pmatrix}$ basis of Column space = 3

$R \begin{pmatrix} 1 & 2 & 2 & -1 \\ 0 & 0 & -1 & 3 \\ 0 & 0 & -1 & 1 \end{pmatrix} \xrightarrow{R_3 - R_2} \begin{pmatrix} 1 & 2 & 2 & -1 \\ 0 & 0 & -1 & 3 \\ 0 & 0 & 0 & -2 \end{pmatrix} \xrightarrow{R_2 \times (-1)} \begin{pmatrix} 1 & 2 & 2 & -1 \\ 0 & 0 & 1 & -3 \\ 0 & 0 & 0 & -2 \end{pmatrix}$

$R \begin{pmatrix} 1 & 2 & 2 & -1 \\ 0 & 0 & 1 & -3 \\ 0 & 0 & 0 & -2 \end{pmatrix} \xrightarrow{R_3 + 2R_2} \begin{pmatrix} 1 & 2 & 2 & -1 \\ 0 & 0 & 1 & -3 \\ 0 & 0 & 0 & 1 \end{pmatrix}$

$\begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 5 \\ 1 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 0 \end{pmatrix}$ Basis of Column space

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3. $x+y+2z=9, 2x+4y-3z=1, 3x+6y-5z=0$

Gaussian elimination

$A = \begin{pmatrix} 1 & 1 & 2 & 9 \\ 2 & 4 & -3 & 1 \\ 3 & 6 & -5 & 0 \end{pmatrix}$

$R \begin{pmatrix} 1 & 1 & 2 & 9 \\ 0 & 2 & -7 & -17 \\ 0 & 3 & -11 & -27 \end{pmatrix} \xrightarrow{R_2 \times \frac{1}{2}, R_3 - \frac{3}{2}R_2} \begin{pmatrix} 1 & 1 & 2 & 9 \\ 0 & 1 & -7/2 & -17/2 \\ 0 & 3 & -11 & -27 \end{pmatrix} \xrightarrow{R_3 - 3R_2} \begin{pmatrix} 1 & 1 & 2 & 9 \\ 0 & 1 & -7/2 & -17/2 \\ 0 & 0 & 1/2 & 3/2 \end{pmatrix}$

$R \begin{pmatrix} 1 & 1 & 2 & 9 \\ 0 & 1 & -7/2 & -17/2 \\ 0 & 0 & 1/2 & 3/2 \end{pmatrix} \xrightarrow{R_2 \times 2, R_3 \times 2} \begin{pmatrix} 1 & 1 & 2 & 9 \\ 0 & 1 & -7/2 & -17/2 \\ 0 & 0 & 1 & 3 \end{pmatrix} \xrightarrow{R_2 + 7/2 R_3} \begin{pmatrix} 1 & 1 & 2 & 9 \\ 0 & 1 & 0 & -11 \\ 0 & 0 & 1 & 3 \end{pmatrix}$

$$z=3$$

$$y - \frac{7}{2}z = -\frac{17}{2} \rightarrow y = -\frac{17}{2} + \frac{7 \cdot 3}{2} = \frac{4}{2} = 2 = y$$

$$x + y + 2z = 9 \Rightarrow x + 2 + 6 = 9 \Rightarrow x = 9 - 8 \Rightarrow x = 1$$

$$x=1, y=2, z=3$$

4. $\begin{bmatrix} 1 & 2 & -3 \\ 0 & -2 & 0 \\ -2 & -2 & 2 \end{bmatrix}$ find inverse. Done

5. $\begin{bmatrix} 1 & 1 & k & 1 & k \\ 0 & -1 & 1 & 2 & 0 \\ 1 & 2 & 1 & -1 & -k \end{bmatrix}$

$$\begin{bmatrix} 1 & 1 & k & 1 & k \\ 0 & -1 & 1 & 2 & 0 \\ 0 & 1 & 1-k & -2 & -2k \end{bmatrix} = \begin{bmatrix} 1 & 1 & k & 1 & k \\ 0 & -1 & 1 & 2 & 0 \\ 0 & 0 & 2-k & 0 & -2k \end{bmatrix}$$

$$2-k = -2k$$

$$2 = -2k + k$$

$$\boxed{-2 = k}$$

PUACP

6. $v_1(1, -2, 3)$, $v_2(5, 6, -1)$, $v_3(3, 2, 1)$ are L.I?

$$a(1, -2, 3) + b(5, 6, -1) + c(3, 2, 1) = (0, 0, 0)$$

$$(a+5b+3c, -2a+6b+2c, 3a-b+c) = (0, 0, 0)$$

$$\begin{bmatrix} 1 & 5 & 3 \\ -2 & 6 & 2 \\ 3 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 5 & 3 \\ 0 & -4 & 8 \\ 0 & -16 & 8 \end{bmatrix} \xrightarrow{R_3 + 4R_2} \begin{bmatrix} 1 & 5 & 3 \\ 0 & -4 & 8 \\ 0 & 0 & 32 \end{bmatrix}$$

L.D

$$7. A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

As it is upper triangular and its eigenvalues are 1, 1, 1

$$P^{-1}AP = D$$

As Eigen vectors are $(1, 0, 0), (0, 1, 0), (0, 0, 1)$

These are L.D so $P = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

$$P^{-1}AP = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

As P is actually identity matrix which means

A is already diagonalizable

2020 MATH-201

(i) $v_1 = (1, -2, 3), v_2 = (5, 6, -1), v_3 = (3, 2, 1)$

$$a(1, -2, 3) + b(5, 6, -1) + c(3, 2, 1) = (0, 0, 0)$$

$$a + 5b + 3c, -2a + 6b + 2c, 3a - b + c = (0, 0, 0)$$

$$\begin{pmatrix} 1 & 5 & 3 \\ -2 & 6 & 2 \\ 3 & -1 & 1 \end{pmatrix} \xrightarrow{R_2 + 2R_1, R_3 - 3R_1} \begin{pmatrix} 1 & 5 & 3 \\ 0 & 16 & 8 \\ 0 & -16 & -8 \end{pmatrix} \xrightarrow{R_3 + R_2} \begin{pmatrix} 1 & 5 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_1 - 5R_2} \begin{pmatrix} 1 & 0 & -2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

L.D

(iii) $(A^T - 2I)^{-1} = \begin{pmatrix} 3 & 0 \\ 1 & -1 \end{pmatrix}$ find A

$$A^T - 2I = \begin{pmatrix} 1/3 & 0 \\ 1/3 & -1/3 \end{pmatrix}$$

$$\text{Det} = -3$$

$$\text{Adj} = \begin{pmatrix} -1 & 0 \\ -1 & 3 \end{pmatrix}$$

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} - 2 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1/3 & 0 \\ 1/3 & -1/3 \end{pmatrix}$$

$$\text{Inverse} = \begin{pmatrix} 1/3 & 0 \\ 1/3 & 1 \end{pmatrix}$$

$$a - 2 = 1/3, b = 0, c = 1/3, d - 2 = -1$$

$$a = 7/3$$

$$d = 1$$

$$\text{So } A = \begin{pmatrix} 7/3 & 0 \\ 1/3 & 1 \end{pmatrix}$$

$$(iii) \quad T(x_1, x_2, x_3) = (-x_2, x_1, x_1 + x_3) \quad \text{Done}$$

$$(iv) \quad v = \{f: \mathbb{R} \rightarrow \mathbb{R}\} \quad W = \{t \in v: f(t) = 0\} \quad \text{Done}$$

$$(v) \quad f(t) = at \cos t, \quad g(t) = b \sin t, \quad h(t) = c \cos t \quad L-17$$

$$a \cos t + b \sin t + c \cos t = 0$$

$$a \cos t + b \sin t + c \cos t = 0$$

$$a \cos t + b \sin t + c \cos t = 0$$

$$a \cos t + b \sin t + c \cos t = 0$$

$$(a+c) \cos t + b \sin t = 0$$

$$t=0 \quad \cos 0 = 1$$

$$t=\pi/2 \quad a+b=0 \quad a+c=0 \Rightarrow a+b+c=0$$

$$t=\pi \quad a-b+c=0$$

$$a+b+c=0 \quad \text{Line 4}$$

$$3. \quad 2x+y+z=1, \quad 3x+y-5z=8, \quad 4x-y+z=5$$

$$\begin{bmatrix} 2 & 1 & 1 & 1 \\ 3 & 1 & -5 & 8 \\ 4 & -1 & 1 & 5 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 0 & 6 & -7 \\ 3 & 1 & -5 & 8 \\ 4 & -1 & 1 & 5 \end{bmatrix} \quad R_1 - R_2 \rightarrow \begin{bmatrix} 1 & 0 & -6 & 7 \\ 3 & 1 & -5 & 8 \\ 4 & -1 & 1 & 5 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & -6 & 7 \\ 0 & 1 & 13 & -13 \\ 0 & -1 & 25 & 23 \end{bmatrix} \quad R_3 + R_2 \rightarrow \begin{bmatrix} 1 & 0 & -6 & 7 \\ 0 & 1 & 13 & -13 \\ 0 & 0 & 38 & 10 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & -6 & 7 \\ 0 & 1 & 13 & -13 \\ 0 & 0 & 1 & \frac{10}{38} \end{bmatrix}$$

$$x = \frac{5}{19}$$

$$y + 13 \left(\frac{5}{19} \right) = -13$$

$$y = -13 - \frac{65}{19}$$

$$y = \frac{312}{19}$$

7.
$$\begin{vmatrix} 1 & 2 & -3 \\ 0 & -2 & 0 \\ -2 & -2 & 2 \end{vmatrix} \text{ Done}$$

5.
$$\begin{vmatrix} 1+x & 1 & 1 & 1 \\ 1 & 1-x & 1 & 1 \\ 1 & 1 & 1+y & 1 \\ 1 & 1 & 1 & 1-y \end{vmatrix} = x^2 y^2$$

6. $v_1(0,0,0), v_2(1,0,1), v_3(1,-1,0), v_4(1,0,-1)$
 $a(0,0,0) + b(1,0,1) + c(1,-1,0) + d(1,0,-1)$

li D

7. Basis of Null space.
$$\begin{bmatrix} 1 & 2 & 2 & -1 \\ 3 & 6 & 5 & 0 \\ 1 & 2 & 1 & 0 \end{bmatrix}$$

$Ax = 0$

$$A = \begin{bmatrix} 1 & 2 & 2 & -1 \\ 3 & 6 & 5 & 0 \\ 1 & 2 & 1 & 0 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 2 & 2 & -1 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & -1 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 2 & -1 \\ 0 & 0 & -1 & 3 \\ 0 & 0 & 0 & -2 \end{bmatrix}$$

$x + 2y + 2z - w = 0$

$0 + 0 - z + 3w = 0$

$z = 3w, x = -2y - 2z + w, y = \text{free variable}$

let $y = w = 1$

$z = 3, y = 1, x = -3$

$N(A) = \{(-3, 1, 3, 1)\}, \text{Dim} = 1$

2021. Math 201

i) $T(-x_3, x_1, x_1 + x_3) = \text{One-to-one}$ Done

ii) $W = \{(x, y) : x, y \geq 0\}$

$W \neq \emptyset$

$u_1 = (1, 2)$ $u_2 = (2, 3)$

$\alpha u_1 + \beta u_2 = \alpha(1, 2) + \beta(2, 3) = (\alpha + 2\beta, 2\alpha + 3\beta) \notin W$

$\alpha u = \alpha(1, 2) \notin W$ W not subspace of $\mathbb{R}^2$

iii) $\begin{bmatrix} 1 & -2 & 3 & -1 \\ 2 & -1 & 2 & 2 \\ 3 & 1 & 2 & 3 \end{bmatrix}$ Reduced echelon form Done

(iv) Show that similar matrices have same eigenvalues

Let A and B be similar matrices, there exist invertible matrix P such that $B = PAP^{-1}$

We want to show A, B have same eigenvalues

Let square A is similar to B , there exist a non-singular matrix P . Consider

$$|B - \lambda I| = |P^{-1}AP - \lambda I|$$

$$= |P^{-1}AP - \lambda P^{-1}P|$$

$$= |P^{-1}(A - \lambda I)P|$$

$$= |P^{-1}| |A - \lambda I| |P|$$

$$= \frac{1}{|P|} |A - \lambda I| |P|$$

$$|B - \lambda I| = |A - \lambda I| \text{ Characteristic Equation of } A \text{ is } A$$

Hence proved

v) Let u be vector in V , assume u is orthogonal to V in V .

$$u \cdot v = 0 \quad \forall v \in V$$

$$v = u \Rightarrow u \cdot u = 0$$

$$\|u\|^2 = 0$$

square of length (= magnitude) of u

Simple method
If $u \in V$ is orthogonal to every $v \in V$

$\langle u, v \rangle = 0 \quad \forall v \in V$
In particular for $v = u$ we have
 $\langle u, u \rangle = 0 \quad \therefore \|u\| = \sqrt{\langle u, u \rangle}$

$$\|u\| = 0 \Rightarrow \|u\| = 0 \Rightarrow u = 0$$

$\|u\| = 0$ u is zero vector

Therefore, if u is orthogonal to every $v \in V$, u must be zero vector.

This result make special sense, as if a vector is orthogonal every other vector in space, it must be "pointing in a direction" that is $\perp$ to all directions which can only be zero vector.

vi) Non similar matrices have same characteristic polynomial

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \quad B = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$$

$$t^2 - 4t + 4 \text{ C.P}$$

$$\lambda = 2, \lambda = 2 \text{ Eigenvalues}$$

A and B are not similar, do not have same eigenvector. A has eigenvector $(1, 0)$ and $(0, 1)$

B has eigenvector $(1, 0)$ and $(1, 1)$

$$\text{or } \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{bmatrix} \Rightarrow (t - 2)(t - 2)^2 = \text{C.P}$$

$$i) \quad x + y + 7z = -7 \quad \text{PUACP}$$

$$2x + 3y + 17z = -16 \quad (\text{Done})$$

$$x + 2y + (a^2 + 1)z = 3a$$

$$ii) \quad \begin{bmatrix} 1 & 2 & -3 \\ 0 & -2 & 0 \\ -2 & -2 & 2 \end{bmatrix} \quad (\text{Done})$$

$$iii) \quad \begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix}$$

$$\begin{aligned} |A - \lambda I| &= \begin{vmatrix} 2-\lambda & 2 & 1 \\ 1 & 3-\lambda & 1 \\ 1 & 2 & 2-\lambda \end{vmatrix} = (2-\lambda)(6-5\lambda+\lambda^2-2) - 2(2-\lambda-1) - 1(2-3\lambda) \\ &= (2-\lambda)(\lambda^2-5\lambda+4) - 2(1-\lambda) - 1(2-3\lambda) \\ &= 2\lambda^2 - 10\lambda + 8 - \lambda^3 + 5\lambda^2 - 4\lambda - 2 + 2\lambda - 2 + 3\lambda \\ &= -\lambda^3 + 7\lambda^2 - 11\lambda + 5 \end{aligned}$$

$$\begin{bmatrix} 1 & -1 & 1 & -5 \\ 1 & 4 & 1 & -6 & 5 \\ 1 & -6 & 5 & 1 \end{bmatrix}$$

$$\lambda - 1 = 0, \lambda^2 - 6\lambda + 5 = 0$$

$$\lambda^2 - 5\lambda - \lambda + 5 = 0$$

$$\lambda(\lambda - 5) - 1(\lambda - 5) = 0$$

$$(\lambda - 5)(\lambda - 1) = 0$$

for $\lambda = 5$,

$$\begin{bmatrix} -3 & 2 & 1 \\ 1 & -2 & 1 \\ 1 & 2 & -3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -2 & 1 \\ -3 & 2 & 1 \\ 1 & 2 & -3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -2 & 1 \\ 0 & -4 & 4 \\ 0 & 4 & -4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -2 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

iv. $2x_1 - x_2 - x_3 = 4$ $3x_1 + 4x_2 = 11$ $3x_1 - 2x_2 + 4x_3 = 11$
 $-2x_3$

$$x_1 - 2x_2 + x_3 = 0 \quad x_1 - x_3 = 0$$

$$x_1 - 2x_3 + x_3 = 0 \quad x_2 = x_3$$

$$x_1 = x_3$$

$$\begin{bmatrix} 1 & 1 & 1 \end{bmatrix}^T$$

for $\lambda = 1$

$$\begin{bmatrix} 1 & 2 & 1 \\ 1 & 4 & 1 \\ 1 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_1 + 2x_2 + x_3$$

$$x_2 = 0, x_3 = b$$

$$x_1 = -2a - b$$

$$\begin{bmatrix} -2a-b \\ a \\ b \end{bmatrix} = \begin{bmatrix} -2a \\ a \\ 0 \end{bmatrix} + \begin{bmatrix} -b \\ 0 \\ b \end{bmatrix} = a \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} + b \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} -2 & 1 & 0 \end{bmatrix}^T, \begin{bmatrix} -1 & 0 & 1 \end{bmatrix}^T$$

(iv)

$$\begin{bmatrix} 2 & -1 & -1 & 4 \\ 3 & 4 & -2 & 11 \\ 3 & -2 & 4 & 11 \end{bmatrix}$$

7.

$$= \begin{bmatrix} 1 & -1/2 & -1/2 & 2 \\ 3 & 4 & -2 & 11 \\ 3 & -2 & 4 & 11 \end{bmatrix} \xrightarrow{\frac{1}{2}R_1}$$

$$= \begin{bmatrix} 1 & -1/2 & -1/2 & 2 \\ 0 & 1/2 & -1/2 & 5 \\ 0 & -1/2 & 11/2 & 5 \end{bmatrix} \xrightarrow{\begin{matrix} R_2 - 3R_1 \\ R_3 - 3R_1 \end{matrix}}$$

$$= \begin{bmatrix} 1 & -1/2 & -1/2 & 2 \\ 0 & 1 & -1/4 & 19/11 \\ 0 & 0 & 60/11 & 63/11 \end{bmatrix} \xrightarrow{R_3 + 11R_2}$$

$$\begin{bmatrix} 1 & -1/2 & 0 & 5/2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix} \xrightarrow{\begin{matrix} R_2 + \frac{1}{2}R_3 \\ R_1 + \frac{1}{2}R_3 \end{matrix}}$$

$$x_1 = 3, x_2 = 1, x_3 = 1$$

(VI) Find equation of subspace spanned by $(1, -3, 5), (-2, 6, -10)$

$$a(1, -3, 5) + b(-2, 6, -10) = (x, y, z)$$

$$a - 2b, -3a + 6b, 5a - 10b = x, y, z$$

$$\begin{bmatrix} 1 & -2 & x \\ -3 & 6 & y \\ 5 & -10 & z \end{bmatrix} \xrightarrow{\begin{matrix} R_2 + 3R_1 \\ R_3 - 5R_1 \end{matrix}}$$

$$x = t, y = -3t, z = 5t \quad \text{parametric equations of straight line}$$

2022

(i) All vector space must be abelian group under multiplication.

from paper

(ii) Show that linear independent set of vectors of any vector space can be extended for basis for V .

let V be vector space and let $S = \{v_1, v_2, \dots, v_n\}$ be linearly independent set of vectors in V

We want to show that S can be extended to basis of V .

1. If $\langle S \rangle$ spans V , S is basis of V
2. If S does not span, there exists a vector v in V that is not span of S . By linear independence of S , the set $S' = \{v_1, v_2, \dots, v_n, v\}$ is still L.I.
3. Repeat 2 step until span S' equal to V . This process must terminate since V is finite dimensional
4. The resulting set S' is basis of V extended original linear I set S . Since S' spans V , it is basis of V . This result is fundamental property of vector space.

3. Let V be V space over field and $W \subseteq V$. Then W is subspace of V or not + example.

i- zero vector of V is in W $(1^2 + 1^2 + 2^2) = 2 \notin S$

ii- for any $u, v \in W$, $u+v \in W$

iii- for any $a \in F$, $u \in W$, $au \in W$

let $V = \mathbb{R}^3$, $W = \{(x, y, 0) \mid x, y \in \mathbb{R}\}$

i- $W \neq \emptyset$

ii- $\alpha(1, 1, 0) + \beta(2, 1, 0) = (\alpha + 2\beta, \alpha + \beta, 0) \in W$ subspace of V

4. Orthogonal:

Similar:

A square matrix is orthogonal if it satisfies

Two matrices A and B are similar if there exists an invertible matrix P such that $B = P^{-1}AP$

1. $Q^T = Q^{-1}$

Similar matrix have properties

2. $QQ^T = Q^TQ = I$

is same det, trace, eigenvalues, rank, nullity

transpose of O.M is its inverse and multiplying matrix by its transpose result in I.M.

Similar matrix have properties is same det, trace, eigenvalues, rank, nullity

It is often used linear transform. It is used to analyze conditions, orthogonal projection linear transformations

A is similar to B

$$\Rightarrow B = P^{-1}AP$$

Multiply $(P^{-1})^n$ from left

$$B(P^{-1})^n(P^n)^n P^{-1}AP = IAP = AP$$

Multiply P^n from right

$$(P^n)^n (AP^{-1})^n = APP^{-1}$$

$$Y^{-1}BX = AI \quad X = P^n$$

$$B = A$$

Hence proved

5. Dependence / Independence vectors:

Let $\{v_1, v_2, \dots, v_n\} \in V$ V is vector space, let $\{a_1, a_2, \dots, a_n\} \in F$ then

$$a_1v_1 + a_2v_2 + \dots + a_nv_n = 0 \quad \forall a_i \in F \quad a_i \neq 0$$

the V is linear dependence otherwise $a_i = 0$

it is said to be L.I.

6. Determinant, cofactor, minors of square square

The det of square matrix A is scalar value

$$\text{denoted by } \det(A) \text{ or } |A|$$

$$A = \begin{bmatrix} a & e \\ f & h \end{bmatrix} = |A| = ah - ef$$

for larger matrices we can use Laplace expansion or other method

The cofactor of element a_{ij} of square matrix A is value obtained by multiplying the minor of a_{ij} by $(-1)^{i+j}$, where

a_{ij} is element with i th row, j th column

It is denoted by C_{ij}

Minor:

PUACP

Minor of element a_{ij} of square matrix is det of submatrix formed by removing i th row and j th column, i.e. If A is 2-by-2 matrix, minor $a_{11} = \det$ of 2x2 submatrix obtained by removing first row and first column

$$\det(AB^T) = 72, \quad \det(A^2B^2) = 144, \quad \det(A), \det(2A), \det(AB^T)$$

$T(x_1, x_2, x_3) = (x_1 - x_2, x_3)$ is it linear find $N(T), R(T)$

$$\text{As } (x_1 - x_2, x_3) = x_1(1, 0) + x_2(-1, 0) + x_3(0, 1)$$

L.I.

$$\text{Now } T(1, 0, 0) = (1, 0) \quad T(0, 0, 1) = (0, 1)$$

$$T(0, 1, 0) = (-1, 0)$$

$$R(T) = \{(1,0), (0,1), (-1,0)\}$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ -1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$$

$(-1,0)$ cast it out

$$R(T) = \{(1,0), (0,1)\} \quad \dim \text{ of } R(T) = 2$$

$$\text{for } N(T) \quad T v = 0$$

$$x_1 - x_2 = 0 \quad x_3 = 0$$

$$x_1 = x_2$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_2 \\ x_2 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

$$N(T) = \{(1,1,0)\} \quad \dim \text{ of } N(T) = 1$$

3. $2x_1 - x_2 - x_3 = 4$, $3x_1 + 4x_2 - 2x_3 = 11$, $3x_1 - 2x_2 + 9x_3 = 11$ Don't

4. Eigen values $\begin{bmatrix} 5 & 9 & 3 \\ -3 & 5 & 6 \\ -1 & -5 & -3 \end{bmatrix}$

$$A - \lambda = \begin{bmatrix} 5-\lambda & 9 & 3 \\ -3 & 5-\lambda & 6 \\ -1 & -5 & -3-\lambda \end{bmatrix} = (5-\lambda)((5-\lambda)(-3-\lambda)+30) - 9(9+3\lambda+6) + 3(15+5-\lambda)$$

$$A - 0 = \begin{bmatrix} 5 & 9 & 3 \\ -3 & 5 & 6 \\ -1 & -5 & -3 \end{bmatrix} = (5-\lambda)(\lambda^2 - 2\lambda + 15) - 135 - 27\lambda + 60 - 3\lambda$$

$$= 5\lambda^2 - 10\lambda + 75 - \lambda^3 + 2\lambda^2 - 19\lambda - 30\lambda - 75$$

$$= -\lambda^3 + 7\lambda^2 - 55\lambda = 0$$

$$-\lambda(\lambda^2 - 7\lambda + 55) = 0$$

$$\lambda = 0$$

$$\lambda^2 - 7\lambda + 55 = 0 \quad \text{complex}$$

$$x_1 = x_2 = x_3$$

$$\begin{bmatrix} 9 & 3 \\ 5 & 6 \end{bmatrix} \quad \begin{bmatrix} 5 & 3 \\ -3 & 6 \end{bmatrix} \quad \begin{bmatrix} 5 & 9 \\ -3 & 5 \end{bmatrix}$$

$$x_1 = x_2 = x_3$$

$$39 \quad 39 \quad 52$$

$$\begin{bmatrix} 39 \\ 39 \\ 52 \end{bmatrix}$$

7. iii) $M_{22} = \begin{bmatrix} 1 & 2 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} 2 & 3 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

$$c_1 \begin{bmatrix} 1 & 2 \\ -1 & 0 \end{bmatrix} + c_2 \begin{bmatrix} 2 & 3 \\ -2 & 1 \end{bmatrix} + c_3 \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$c_1 + 2c_2 + c_3 a = 0 \quad 2c_1 + 3c_2 + bc_3 = 0 \quad -c_1 - 2c_2 + c_3 c = 0$$

$$c_2 + dc_3 = 0$$

$$\begin{bmatrix} 1 & 2 & a \\ 2 & 3 & b \\ -1 & -2 & c \\ 0 & 1 & d \end{bmatrix}$$

The last matrix has $R=2$ if:

$$c+a=0 \quad b+d-2a=0$$

The required conditions on a, b, c, d

$$a+c=0 \quad b+d-2a=0$$

$$\text{for } a=b=d=1, c=-1$$

$$c_1 - c_3 = 1, c_2 = -1$$

2022.

i) If $|A| \neq 0$ $|A^t| \neq 0$ $(A^t)^{-1} = (A^{-1})^t$

Given that,

$$|A| \neq 0 \quad |A^t| \neq 0$$

n is non-zero integer

$$(A^t)^{-1} = (A^{-1})^t$$

A is nonsingular

$$AA^{-1} = I$$

$$(AA^{-1})^t = I^t$$

$$(A^{-1})^t A^t = I$$

Multiply $(A^t)^{-1}$

$$(A^{-1})^t A^t (A^t)^{-1} = I (A^t)^{-1}$$

$$(A^{-1})^t I = (A^t)^{-1}$$

$$(A^{-1})^t = (A^t)^{-1} \quad \text{proved}$$

ii) For square matrix A satisfies equation $A^2 + 2A + I = 0$

$$|A| \neq 0$$

Given that,

$$A^T A = I_3$$

$$\text{rank } A^T = 3$$

$$A^T A = I_3 \Rightarrow A^T = I_3 A$$

$$A^T A = I_3 \Rightarrow A^T = I_3 A$$

$$A^T = I_3 A$$

$$A^T = A^{-1}$$

$$\det(A) \neq 0$$

iii) Using row operation

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 5 & 3 & 1 & 0 & 1 \\ 0 & 1 & -3 & 2 & 1 & 0 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{bmatrix}$$

وقل رب زدني علما

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 0 & 0 & 0 \\ 0 & 1 & -3 & 2 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 5 & -2 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{matrix} R_1 - 2R_2 \\ R_2 + 3R_3 \\ R_3 - 5R_4 \end{matrix}$$

PUACP

$$\begin{bmatrix} 1 & 0 & 0 & -40 & 16 & 9 \\ 0 & 1 & 0 & 13 & 5 & -3 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{bmatrix} \begin{matrix} R_1 + 40R_3 \\ R_2 - 13R_3 \\ R_1 - 16R_3 \end{matrix}$$

iv) Domain, range of $T(x_1, x_2, x_3, x_4) = (y_1, y_2)$

$$\text{let } U = (x_1, x_2, x_3, x_4) \quad V = (y_1, y_2, y_3, y_4)$$

$$T_U = x_1, x_2$$

$$T_V = y_1, y_2$$

We can see domain of T is R^4 and range of T is R^2

v) Area of Δ $(3, 3), (4, 0), (-2, -1)$

$$= \frac{1}{2} (x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2))$$

$$= \frac{1}{2} | 3(0 + 1) + 4(-1 - 3) + (-2)(3 - 0) | = \frac{1}{2} | 3 - 16 - 6 | = \frac{1}{2} | -19 | = \frac{19}{2}$$

1. We Show $u = (-2, 3, 1, 4)$, $v = (1, 2, 0, -1)$ orthogonal vectors of $\mathbb{R}^4$. Also find angle b/w u and v .
 u and v are orthogonal if $\langle u, v \rangle = 0$
 $\langle u, v \rangle = -2 + 6 - 4 = 0$
 $\langle u, v \rangle = 0$

So u is orthogonal to v and angle is 90° .

Q. If A is invertible iff not eigen value of A .

Let A be square matrix of order n .
 Suppose that A is invertible i.e. A^{-1} exists

$$\det A \neq 0$$

$$\det(A - \lambda I) \neq 0$$

$$\det(A - \lambda I) \neq 0$$

$\lambda = 0$ is not a root of characteristic equation hence $\lambda = 0$ is not an eigen value of A .

i. Conversely suppose $\lambda = 0$ is not an eigen value of A .

λ is not root of characteristic equation

$$\det(A - \lambda I) \neq 0$$

$$\det(0I - A) \neq 0$$

$$\det(-A) \neq 0$$

$$(-1)^n \det A \neq 0$$

$$\det A \neq 0$$

A is an invertible matrix

Prove

2. $T(x, y, z) = (x, z, x+y)$ is a linear transformation

Find $R(T)$ and $N(T)$

$$T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$$

$$T(1, 0, 0) = (1, 0, 1)$$

$$T(0, 1, 0) = (0, 0, 1)$$

$$T(0, 0, 1) = (0, 1, 0)$$

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$R(T) = \{(1, 0, 1), (0, 0, 1), (0, 1, 0)\}$$

(Dimension of $R(T)$) = 3

Now $N(T) = \{(x, y, z) \in \mathbb{R}^3 \mid T(x, y, z) = 0\}$

$$\begin{matrix} x+y \\ z=0 \end{matrix}$$

$$y=0$$

$$(0, 0, 0) \in N(T) \quad (\text{Dimension of } N(T) = 0)$$

4. $(1, 1, -4, -3), (2, 0, 2, -2), (2, -1, 3, 2)$

$$a(1, 1, -4, -3) + b(2, 0, 2, -2) + c(2, -1, 3, 2)$$

$$(a+2b+2c, a-b, -4a+2b+3c, -3a-2b+2c)$$

$$a+2b+2c=0$$

$$a-b=0$$

$$-4a+2b+3c=0$$

$$a=c$$

$$-3a-2b+2c=0$$

$$a=0$$

$$2b+3c=0$$

$$b=0$$

$$2b+c=0$$

$$4c=0$$

$$c=0$$

HL dimension is 3